

Def. A continuous real-valued $f: D \rightarrow \mathbb{R}$ on a domain $D \subseteq \mathbb{C}$ is called harmonic if:
 f is partially twice continuously differentiable ($f \in C^2$) satisfying

$$f_{xx} + f_{yy} = 0. \quad (\text{It is called the Laplace's equation.})$$

Thm. Let $f = u + iv$ be an holomorphic on a domain D .

Then, both u and v are harmonic on D .

Proof. By the Cauchy-Riemann equation, $u_x = v_y$ and $u_y = -v_x$, so that

$$f' = u_x + i v_x = u_x - i \cdot u_y = v_y + i \cdot v_x$$

$$\begin{aligned} \hookrightarrow f'' &= u_{xx} - i u_{xy} = v_{yx} + i v_{xx} \\ &= u_{xx} - i u_{xy} = v_{yx} + i v_{xx} \\ &= u_{xx} - i v_{yy} = -u_{yy} + i v_{xx} \end{aligned} \quad \begin{array}{l} \\ \\ \Downarrow \text{Fubini's Theorem} \end{array}$$

Therefore, $u_{xx} = -u_{yy}$ and $-v_{yy} = v_{xx}$.

Thm.

If $u: D \rightarrow \mathbb{R}$ is harmonic on a simply connected domain D , then there exists an analytic function f on D satisfying $\operatorname{Re}(f) = u$ on D .

Proof. Define $g(z) = u_x(z) - i \cdot u_y(z)$. Then, since the u is harmonic and so $u \in C^2$,

$$u_{xx} = -u_{yy}, \quad \text{and} \quad u_{xy} = u_{yx} \quad \text{by the Fubini's Theorem,} \quad \dots (*)$$

Actually, the $(*)$ is a Cauchy-Riemann equation for g , so g is analytic on D , since $u \in C^2$. Let $z_0 \in D$ be fixed. Define

$$F(z) = \int_{z_0}^z g(\mu) d\mu \quad \text{on } D.$$

So F is defined as integral of an analytic function, so F is analytic.